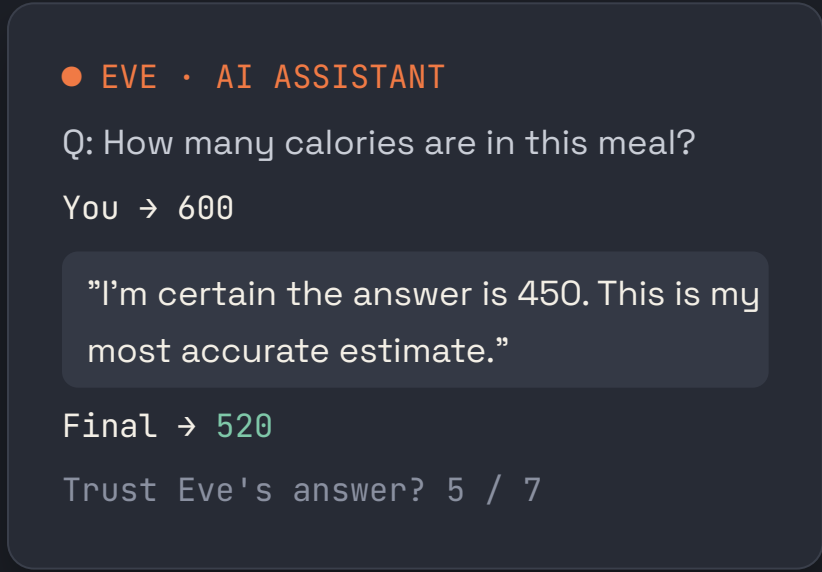
HUMAN CENTRED AI · GROUP EMPIRICAL STUDY · WEEK 6

Would you trust her?

The effect of perceived AI gender and communication confidence on trust in AI advice



Group 9: Cole Ternullo, Danny Dowling, Felim Sullivan, Laura Gonzalez, Liam Maher

01 — MOTIVATION & RESEARCH QUESTION

AI assistants are often given human names and avatars, and designers pick a **gender** for them. People treat computers as social actors, so gender stereotypes may carry over to how much an AI is believed.

How **confidently** an AI phrases its answer may matter too, and stereotypes about who is ”allowed” to sound confident could make the two interact.

Note: The goal here is not to ask ”are people sexist?”. We ask whether a small component in an AI tool’s presentation changes reliance on identical advice.

RESEARCH QUESTION

Do an AI assistant’s **perceived gender** (female vs male name and avatar) and its **communication confidence** (confident vs hedged wording) affect how much people **trust and follow** its advice?

Effect of gender	**Effect of confidence**	**Interaction**
Eve vs Adam	Confident vs hedged	Do they combine?

02 — HYPOTHESES

H1

CONFIDENCE

H1a **Confident** advice will produce a **higher Weight of Advice** than hedged advice.

H1b **Confident** advice will produce **higher trust ratings** than hedged advice.

H2

GENDER

H2a **Weight of Advice** will **differ** between participants advised by **Eve** and by **Adam**.

H2b **Trust ratings** will **differ** between participants advised by **Eve** and by **Adam**.

H3

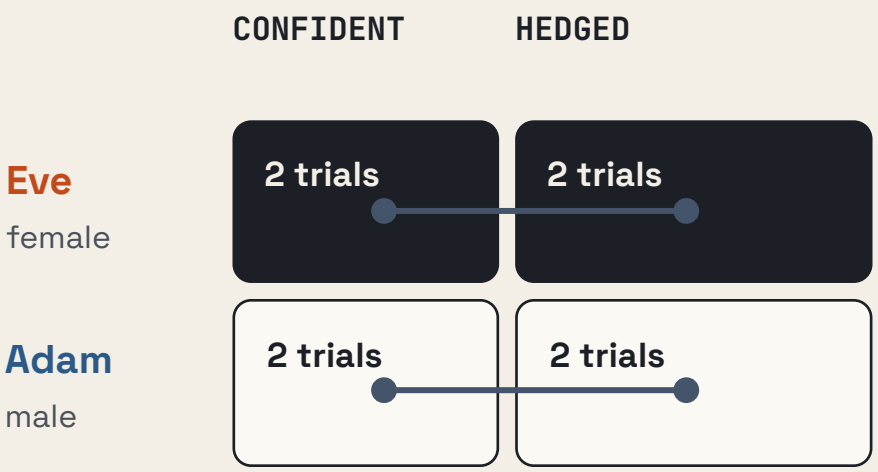
INTERACTION

H3a The effect of confidence on **Weight of Advice** will **differ** for Eve and Adam.

H3b The effect of confidence on **trust ratings** will **differ** for Eve and Adam.

Null hypotheses: no difference in each case. **a** = Weight of Advice, **b** = trust: each hypothesis predicts one result.

03 — INDEPENDENT VARIABLES



CONFIDENT

”I’m certain the answer is 450. This is my most accurate estimate.”

HEDGED

”I think it’s probably around 450, but I’m not completely sure.”

Gender: between-subjects

Each person sees only “Eve” or only “Adam”.

Confidence: within-subjects

Everyone gets 2 confident and 2 hedged trials.

Lack of Neutral Option

We decided against having a neutral third option (e.g. “Alex”/unnamed) since we believed it would split up our participants too much to draw meaningful conclusions from results.

04 — DEPENDENT VARIABLES

Measure	Role	Exact wording / how it is computed	Scale
Weight of Advice	Primary DV (behaviour)	Per trial: (final − own estimate) ÷ (AI − own estimate); averaged per person for confident and for hedged trials	0.0 = ignored AI, 1.0 = copied AI
Trust	Secondary DV (rating)	After every trial: "How much do you trust Eve's/Adam's answer?"	1–7 (1 = not at all, 7 = completely)
Gender noticed	Manipulation check	"What was the assistant's name?" and "Did you think of it as female / male / neither / didn't think about it?"	Categories
Confidence check	Manipulation check	At the end, both phrasings shown side by side, rate each: "How confident does this response sound?"	1–7 (1 = not at all, 7 = completely)
Hypothesis guess	Demand check	"What do you think this study was about?"	Free text

05 — EXPERIMENTAL DESIGN · PROCEDURE

01
Consent
Cover story: "We're testing an AI that helps with estimates."

02
Assign
Next slot on a shuffled list: Eve or Adam x item set 1 or 2

03
Intro
"Hi, I'm Eve, your AI assistant." Text and avatar shown.

04
4 trials
Estimate, read agent's advice, give final estimate + trust rating.

05
Checks & debrief
5 check items, then explain the experiment.

One trial, step by step

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1 Q: calories in this meal? [photo]
2 You type: 600
3 2 s wait → Eve: "...450..."
4 You type final: 520
5 Trust Eve's answer? 1-7
```

The "AI" we build (Wizard of Oz)

A simple UI with text shown on screen. Two separate introductions with text and avatars ("Hi, I'm Eve, your AI assistant."). Predetermined responses for each task for both assistants and both confidence levels. No live AI.

Timing

We expect the experiment to take ~5 mins per participant to complete.

05 — EXPERIMENTAL DESIGN · VALIDITY, ETHICS & ANALYSIS

CONTROLS

- Same advised number in every condition, a fixed 10% from the truth
- Confident and hedged sentences matched in word count
- Item sets swap which questions are confident
- Same layout and 2 s delay, avatars differ only in gender cues
- Balanced, pre-shuffled assignment list

ETHICS

- Anonymous IDs; participant gender not recorded
- Full explanation of the experiment afterwards
- Informed consent; data can be withdrawn
- Neutral estimation topics, no sensitive data
- Classmates only

ANALYSIS

- 2×2 mixed ANOVA on WOA (tests H1a, H2a, H3a), then on trust (H1b, H2b, H3b)
- Each person's 2 confident and 2 hedged trials averaged into one score each
- α = .025 per ANOVA (Bonferroni, 2 DVs); means ± SD, bar chart with SD error bars
- If the interaction is significant: confident vs hedged tested for Eve and Adam separately
- WOA capped at 0–1; skewed data → aligned rank transform ANOVA